Performance Landscapes in Text Classification

How feature representation and regularization shape logistic regression performance

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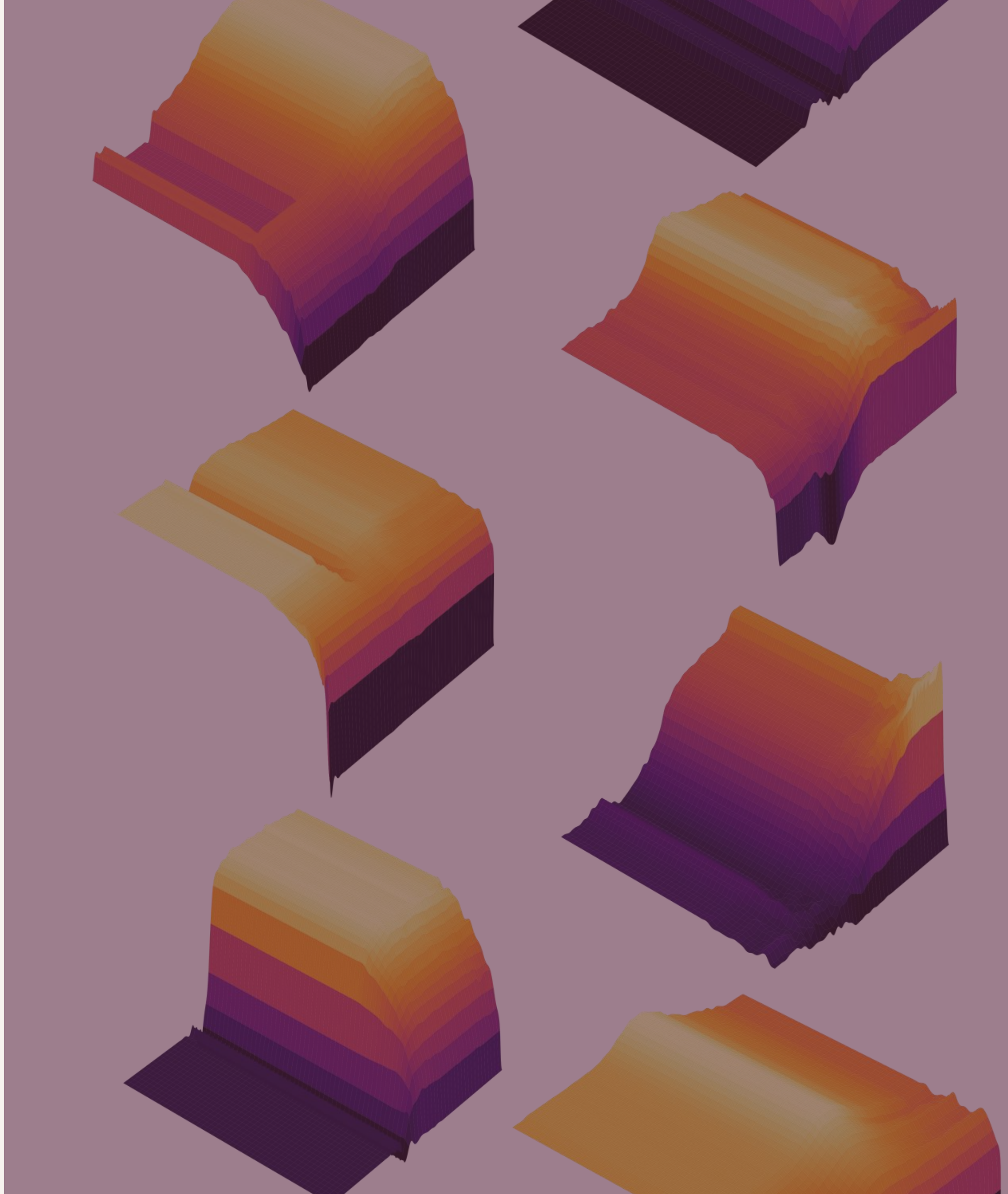


SETUP

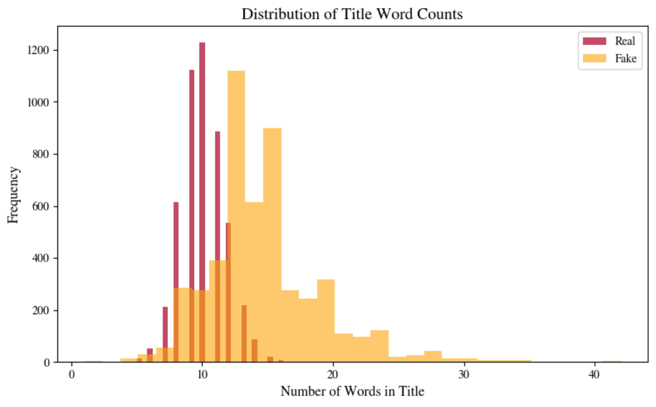
Data and context

Specifications

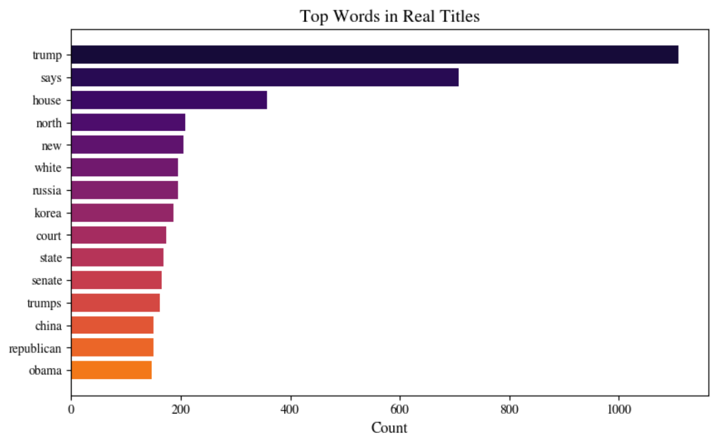
- 10,000 titles, balanced 50/50 fake vs. real
- Real articles from Reuters; fake articles from flagged unreliable sources
- Title-only subset made the grid feasible to run
- Short headlines make representation effects easier to interpret

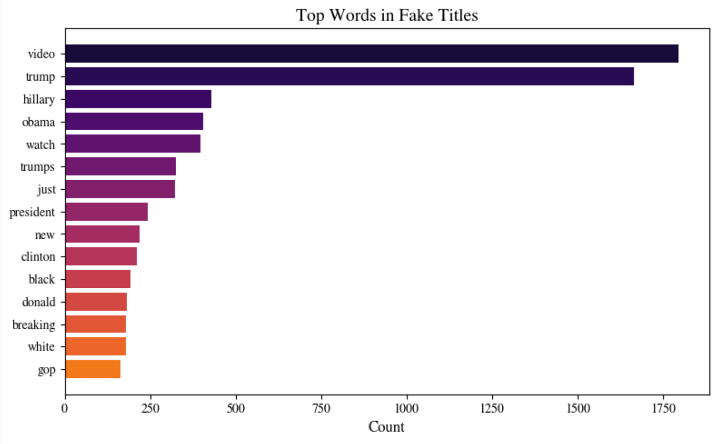
Question: where does performance fail, level off, or become fragile?

Title lengths



Common words by class





Real titles lean more institutional/geopolitical. Fake titles use more attention-grabbing terms.

METHOD

How the study works

Core idea: map the whole parameter space, not just the single best score.

Titles → **Features BoW / TF-IDF uni / uni+bi / bi** → **Logistic regression L1 or L2** → **Heatmaps over C × max_features read with F1, AUC, gap**

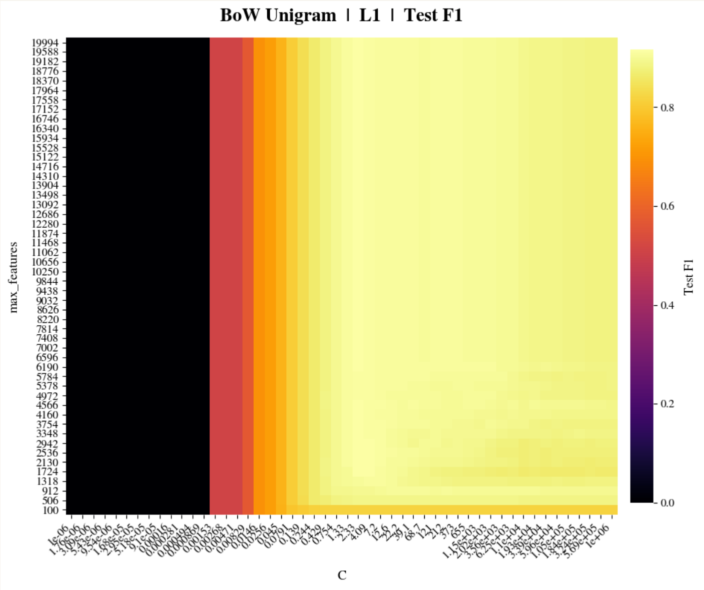
- 80/20 stratified train/test split
- 36 heatmaps total: 2 representations × 3 n-gram settings × 2 penalties × 3 metrics
- Goal: compare broad structures, not just one best cell
- Key idea: stable plateau > isolated peak

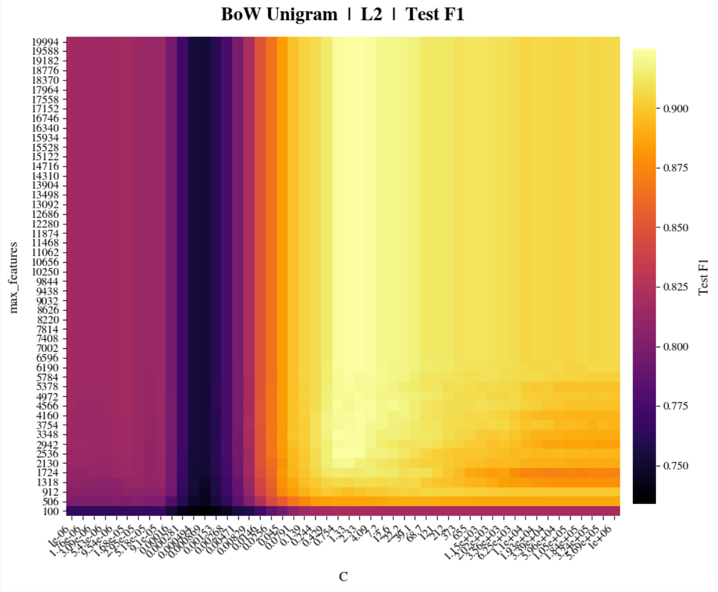
Motivation
A best score alone does not show whether a setting is part of a broad stable region or a narrow fragile one.

RESULT

Finding 1: The penalty changes the regime

The same model behaves very differently depending on the regularization penalty.





What changes?

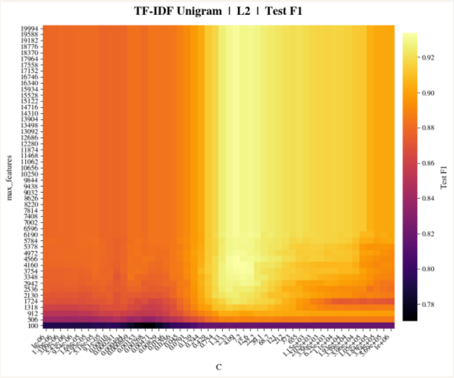
- L1 creates a sharp low-C failure zone
- L1 also has a lower score ceiling
- L2 weakens more gradually instead of collapsing outright
- So the penalty changes the shape of the landscape, not just the score

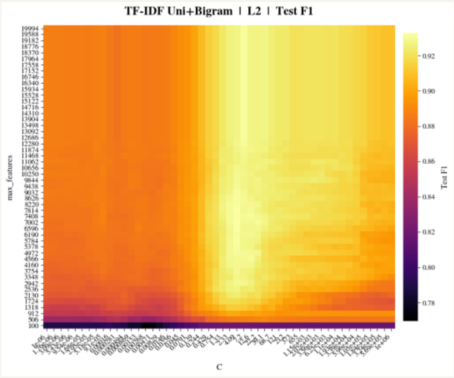
Penalty choice changes the landscape first

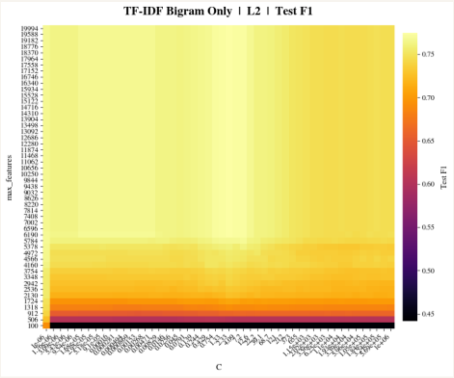
RESULT

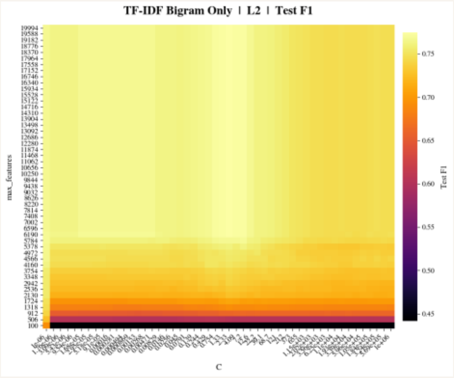
Finding 2: Most of the signal is word-level

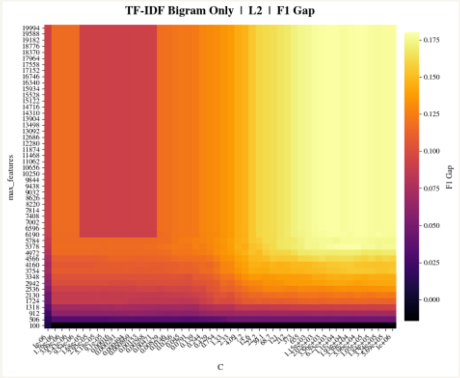
Under TF-IDF + L2, unigram and unigram + bigram look almost identical; bigram-only is the one that breaks away.











Bigram-only also shows larger train-test gaps.

Interpretation

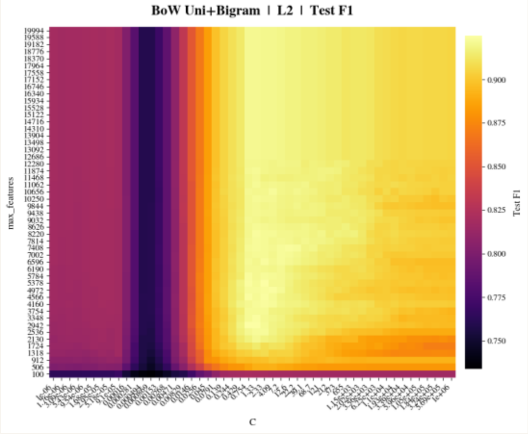
- Unigrams already capture most of the useful signal in short titles
- Adding bigrams changes little; removing unigrams hurts a lot
- Bigram-only needs more features (~5,000 – 6,000) and still stays weaker and less stable

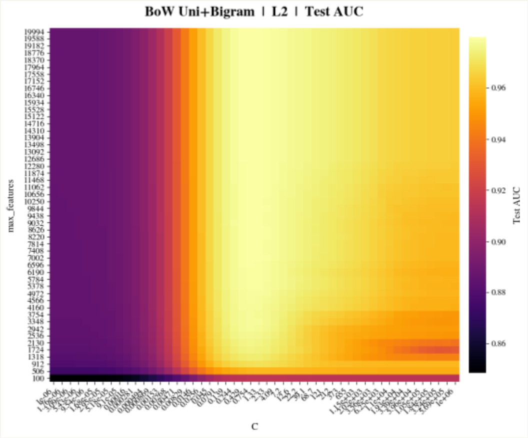
Practical implication: keep unigram information.

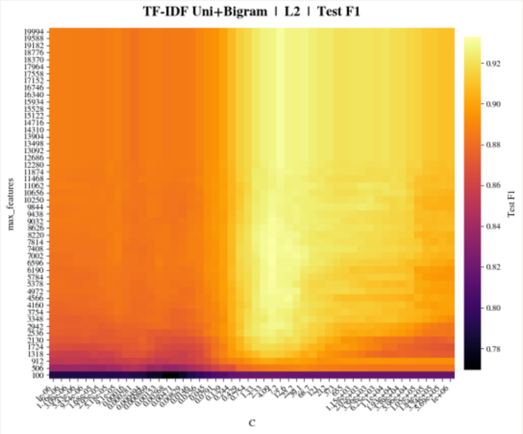
RESULT

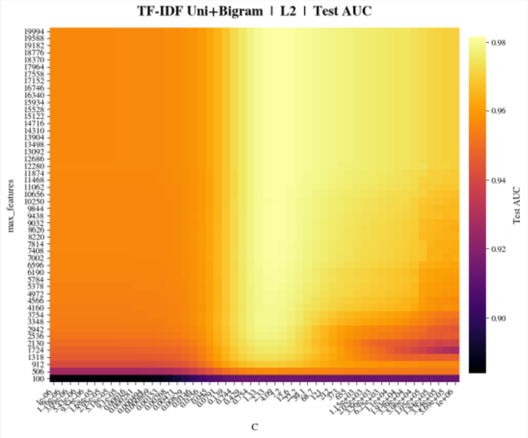
Finding 3: Metric choice changes what you see

The BoW + L2 valley is the key example: it appears in F1, disappears in AUC, and is much less visible in TF-IDF.









Interpretation

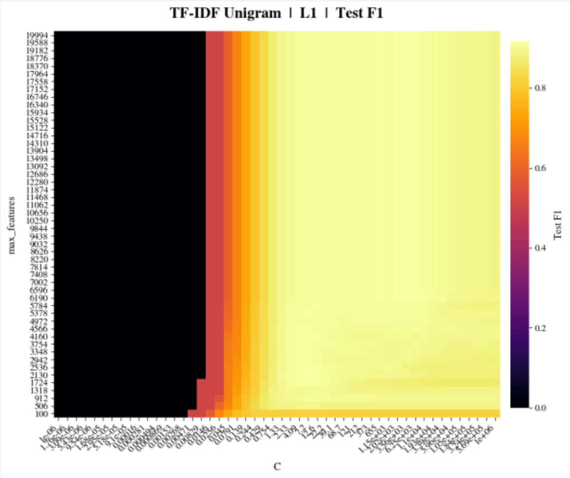
- Logistic regression outputs scores first, then class labels
- AUC is measures ranking; F1 measures final classifications
- In BoW + L2, the model seems to keep ranking fairly well while becoming less decisive near the cutoff
- TF-IDF weakens that effect by downweighting common words
- The TF-IDF comparison suggests this valley is tied to BoW + L2, not just L2 alone
- That is why using both metrics matters.

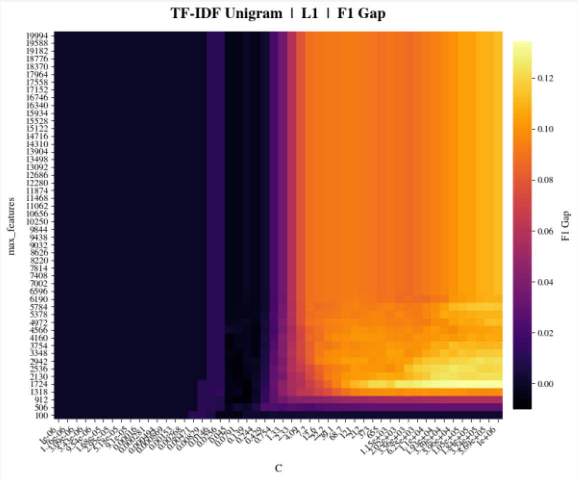
F1 alone would make this region look like a full failure.

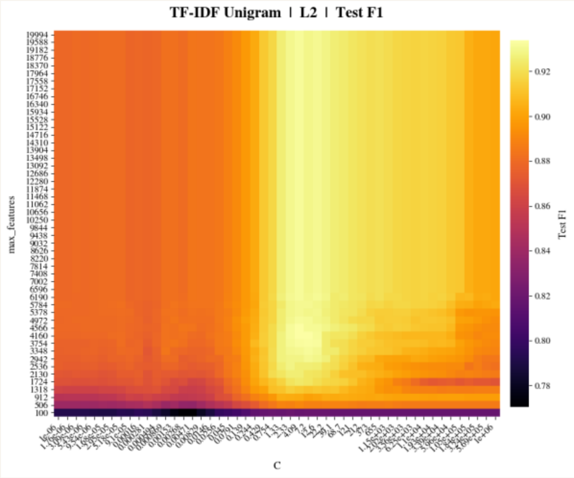
RESULT

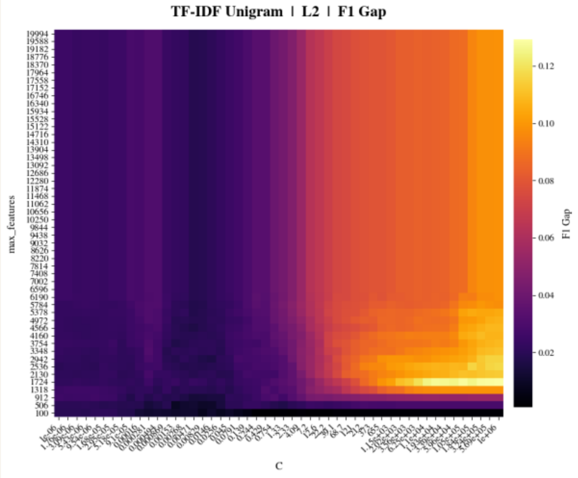
Finding 4: The best score is not always the best region

Train-test F1 gap adds a second filter: strong test performance is not enough by itself









Interpretation

- F1 rises quickly and then plateaus, but the gap keeps increasing as C gets larger
- The weakest-regularization extreme often adds little performance while making the model less stable
- The most useful region is usually a **moderate-C plateau**: performance is already near its best, but the gap is still smaller

Choose settings using performance + stability, not score alone

CONCLUSION

Key Takeaways

- L1 and L2 create different performance regimes
- In short titles, most useful signal is already captured by unigrams
- F1 and AUC reveal different structures, so one metric is not enough
- Default for similar short-text datasets: unigram-inclusive features, about 2,000 to 3,000 max features, and search the moderate-C plateau before pushing to the weakest-regularization extreme

Map the landscape, don't just chase the best cell

Questions?